

	First version
	General notation $1_c(x)$ indicator function: 1 if $x \in C$, else 0
Sets and parameters: D set of all days of the scheduling period; D is an ordered set of elements such that $d = 1$ is the first day and $d = D $ is the last day of the scheduling period. T set of possible timeslots in any day; T is an ordered set of elements such that $t = 1$ is the first timeslot of the day and $t = T $ is the last time slot of the day. T_d set of timeslots $t \in T$, $T_d \subset T$, in day $d \in D$ W set of all workers $w \in W$ S set of all skills $s \in S$ S_w set of skills $s \in S$, $S_w \subset S$, that can be assigned to worker $w \in W$ H_{wd} set of all possible daily working assignments for worker $w \in W$ in day $d \in D$; each assignment $h \in H_{wd}$ is defined by the binary parameters δ_{wdht} indicating if timeslot $t \in T_d$ is covered by the daily working assignment $h \in H_{wd}$ for worker $w \in W$ in day $d \in D$ α_{dts} minimum number of workers required on day $d \in D$, timeslot $t \in T_d$ and skill $s \in S$	Sets and parameters D ordered set of all days of the scheduling period, where $d = 1$ is the first day and $d = D $ is the last day of the scheduling period T ordered set of all possible timeslots in any day, where $t = 1$ is the first timeslot of the day and $t = T $ is the last time slot of the day $T_d \subset T$ set of timeslots that must be covered in day $d \in D$ W set of all workers $w \in W$ S set of all skills $s \in S$ $S_w \subset S$ set of skills of a worker $w \in W$ H_{wd} set of all possible schedules $h \subset T_d$ for worker $w \in W$ in day $d \in D$ α_{dts} minimum number of workers required on day $d \in D$, timeslot $t \in T_d$ and skill $s \in S$
Variables: x_{wdh} binary variable that, when equal to 1, indicates that worker $w \in W$ works in day $d \in D$ in daily working assignments $h \in H_{wd}$ y_{wdts} binary variable that, when equal to 1, indicates that worker $w \in W$ works in day $d \in D$ in time slot $t \in T$ with skill $s \in S$	Variables x_{wdh} binary variable that, when equal to 1, indicates that worker $w \in W$ works in day $d \in D$ in day schedule $h \in H_{wd}$ y_{wdts} binary variable that, when equal to 1, indicates that worker $w \in W$ works in day $d \in D$ in time slot $t \in T$ with skill $s \in S$

z_{dts} integer variable with the number of workers below the required value α_{dts} on day $d \in D$, timeslot $t \in T_d$ and skill $s \in S$	z_{dts} integer variable with the number of workers missing to reach the required value α_{dts} on day $d \in D$, timeslot $t \in T_d$ and skill $s \in S$
A possible objective function: $\text{Minimize } \sum_{w \in W} \sum_{t \in T_d} \sum_{s \in S} z_{dts} \quad (1)$ With this function, the aim is to minimize the sum of the number of workers below the required value for all days $d \in D$, timeslots $t \in T_d$ and skills $s \in S$. NOTA: variável do primeiro somatório não está bem	A possible objective function to minimize $\sum_{d \in D} \sum_{t \in T_d} \sum_{s \in S} z_{dts} \quad (1)$ With this function, the aim is to minimize the sum of the number of workers below the required value for all days $d \in D$, timeslots $t \in T_d$ and skills $s \in S$.
Constraints: $\sum_{h \in H_{wd}} x_{wdh} \leq 1, w \in W, d \in D \quad (2)$ With constraints (2), each worker $w \in W$ can be assigned on each day $d \in D$ with at most one working daily assignment $h \in H_{wd}$. $\sum_{s \in S_w} y_{wdts} = \sum_{h \in H_{wd}} \delta_{wdht} x_{wdh}, w \in W, d \in D, t \in T_d \quad (3)$ With constraints (3), each worker $w \in W$ assigned on day $d \in D$ with a working daily assignment $h \in H_{wd}$ that covers timeslot $t \in T_d$ must be assigned in timeslot t with one of its skills S_w. $\sum_{d'=d}^{d+5} \sum_{h \in H_{wd'}} x_{wd'h} \leq 5, w \in W, d = 1, 2, \dots, D - 5 \quad (4)$ With constraints (4), each worker $w \in W$ cannot be assigned with more than 5 working days in any set of 6 consecutive days. $z_{dts} + \sum_{w \in W} y_{wdts} \geq \alpha_{dts}, d \in D, t \in T_d, s \in S \quad (5)$	Constraints $\sum_{h \in H_{wd}} x_{wdh} \leq 1, w \in W, d \in D \quad (2)$ With constraints (2), each worker $w \in W$ can be assigned on each day $d \in D$ with at most one working daily assignment $h \in H_{wd}$. $\sum_{s \in S_w} y_{wdts} = \sum_{h \in H_{wd}} 1(t \in h) x_{wdh}, w \in W, d \in D, t \in T_d \quad (3)$ With constraints (3), each worker $w \in W$ assigned on day $d \in D$ with a working daily assignment $h \in H_{wd}$ that covers timeslot $t \in T_d$ must be assigned in timeslot t with one of its skills S_w. $\sum_{d'=d}^{d+5} \sum_{h \in H_{wd'}} x_{wd'h} \leq 5, w \in W, d = 1, 2, \dots, D - 5 \quad (4)$ With constraints (4), each worker $w \in W$ cannot be assigned with more than 5 working days in any set of 6 consecutive days. $z_{dts} + \sum_{w \in W} y_{wdts} \geq \alpha_{dts}, d \in D, t \in T_d, s \in S \quad (5)$

With constraints (5), the value of variable z_{dts} is at least the number of workers below the required value α_{dts} for each day $d \in D$, each timeslot $t \in T_d$ and each skill $s \in S$.	With constraints (5), the value of variable z_{dts} is at least the number of workers below the required value α_{dts} for each day $d \in D$, each timeslot $t \in T_d$ and each skill $s \in S$.
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Second version (adding competences of workers on each of their skills)

NOTE: This version is required only if the objective function is required to take into consideration the competences of the workers. For illustration purposes, I'm assuming one objective function that is defined with the competences. The additions to the previous model are highlighted in yellow.

Sets and parameters:

D	set of all days of the scheduling period; D is an ordered set of elements such that $d = 1$ is the first day and $d = D $ is the last day of the scheduling period.
T	set of possible timeslots in any day; T is an ordered set of elements such that $t = 1$ is the first timeslot of the day and $t = T $ is the last time slot of the day.
T_d	set of timeslots $t \in T$, $T_d \subset T$, in day $d \in D$
W	set of all workers $w \in W$
S	set of all skills $s \in S$
C	set of all competences $c \in C$; C is an ordered set of elements such that $c = 1$ represents the highest competence and $c = C $ represents the lowest competence
S_w	set of skills $s \in S$, $S_w \subset S$, that can be assigned to worker $w \in W$; for each $s \in S_w$, the parameter $c_{ws} \in C$ indicates the competence of worker $w \in W$ in skill $s \in S_w$
H_{wd}	set of all possible daily working assignments for worker $w \in W$ in day $d \in D$; each assignment $h \in H_{wd}$ is defined by the binary parameters δ_{wdht} indicating if timeslot $t \in T_d$ is covered by the daily working assignment $h \in H_{wd}$ for worker $w \in W$ in day $d \in D$

α_{dts} minimum number of workers required on day $d \in D$, timeslot $t \in T_d$ and skill $s \in S$

Variables:

x_{wdh} binary variable that, when equal to 1, indicates that worker $w \in W$ works in day $d \in D$ in daily working assignments $h \in H_{wd}$

y_{wdts} binary variable that, when equal to 1, indicates that worker $w \in W$ works in day $d \in D$ in time slot $t \in T$ with skill $s \in S$

y'_{wdtsc} binary variable that, when equal to 1, indicates that worker $w \in W$ works in day $d \in D$ in time slot $t \in T$ with skill $s \in S$ and competence $c \in C$

z_{dts} integer variable with the number of workers below the required value α_{dts} on day $d \in D$, timeslot $t \in T_d$ and skill $s \in S$

z'_{dts} binary variable that is 1 if the number of workers with the highest competence $c = 1$ on day $d \in D$, timeslot $t \in T_d$ and skill $s \in S$ is below one.

A possible objective function:

$$\text{Minimize } \sum_{w \in W} \sum_{t \in T_d} \sum_{s \in S} z_{dts} \quad (1)$$

With this function, the aim is to minimize the sum of the number of workers below the required value for all days $d \in D$, timeslots $t \in T_d$ and skills $s \in S$.

Another possible objective function:

$$\text{Minimize } \sum_{w \in W} \sum_{t \in T_d} \sum_{s \in S} z'_{dts} \quad (1')$$

With this function, the aim is to minimize the sum of the number days $d \in D$, timeslots $t \in T_d$ and skills $s \in S$ that do not have at least one worker with competence $c = 1$.

Constraints:

$$\sum_{h \in H_{wd}} x_{wdh} \leq 1, \quad w \in W, d \in D \quad (2)$$

With constraints (2), each worker $w \in W$ can be assigned on each day $d \in D$ with at most one working daily assignment $h \in H_{wd}$.

$$\sum_{s \in S_w} y_{wdts} = \sum_{h \in H_{wd}} \delta_{wdht} x_{wdh} \quad , w \in W, d \in D, t \in T_d \quad (3)$$

With constraints (3), each worker $w \in W$ assigned on day $d \in D$ with a working daily assignment $h \in H_{wd}$ that covers timeslot $t \in T_d$ must be assigned in timeslot t with one of its skills S_w .

$$\sum_{d=d}^{d+5} \sum_{h \in H_{wd}} x_{wdh} \leq 5 \quad , w \in W, d = 1, 2, \dots, |D| - 5 \quad (4)$$

With constraints (4), each worker $w \in W$ cannot be assigned with more than 5 working days in any set of 6 consecutive days.

$$z_{dts} + \sum_{w \in W} y_{wdts} \geq \alpha_{dts} \quad , d \in D, t \in T_d, s \in S \quad (5)$$

With constraints (5), the value of variable z_{dts} is at least the number of workers below the required value α_{dts} for each day $d \in D$, each timeslot $t \in T_d$ and each skill $s \in S$.

$$y_{wdts} = y'_{wdtsc} \quad , w \in W, d \in D, t \in T_d, s \in S_w, c = c_{ws} \quad (6)$$

With constraints (6), each worker $w \in W$ that works in day $d \in D$ in time slot $t \in T$ with skill $s \in S$ is assigned with its competence c_{ws} in skill s .

$$z'_{dts} + \sum_{w \in W} y'_{wdtsc} \geq 1 \quad , d \in D, t \in T_d, s \in S, c = 1 \quad (7)$$

With constraints (7), the value of variable z'_{dts} is one if the number of workers with competence $c = 1$ for each day $d \in D$, each timeslot $t \in T_d$ and each skill $s \in S$ is zero.